

Technical Document #967: Retrieval Augmented Generation - Comprehensive Overview

Technical Document #967: Retrieval Augmented Generation - Comprehensive Overview

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Re-ranking models score retrieved documents for relevance to the query.
- Hybrid search combines keyword-based and semantic search.